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25 August 1986
SUPERSEDING
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6 December 1954
(See 6,11)

MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, CARGO OIL SERVICE NAVAL SHIPBOARD-USE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements applicable to the design and construction of centrifugal pumps for cargo oil service and sea water ballast.

1.2 Classification. Centrifugal cargo oil service pumps covered by this specification shall be of the following type (see 6.2.1):

Type I - Horizontal, split case
Type II - Vertical deep well turbine

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-B-187 - Bearings, Roller, Tapered.
QQ-C-390 - Copper Alloy Castings (Including Cast Bar).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL (Continued)

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought.
QQ-N-288	-	Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
TT-P-645	-	Primer, Paint, Zinc Chromate, Alkyd Type.
WW-P-404	-	Pipe, Steel, (Seamless and Welded, Black and Zinc-coated (Galvanized)).
GGG-P-781	-	Pullers, Bushings, Bearing, Gear, Wheel Hub and Cylinder Sleeve Installing and Removing.

MILITARY

MIL-S-901	-	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirement for.
DOD-D-1000	-	Drawings, Engineering and Associated Lists.
MIL-S-1222	-	Studs, Bolts, Hex Cap Screws, and Nuts.
MIL-C-2212	-	Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices,
MIL-P-15024	-	Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	-	Plates, Identification.
MIL-S-15083	-	Steel Castings.
MIL-E-15090	-	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-P-15137	-	Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
DOD-P-15328	-	Primer (Wash) Pretreatment (Formula No, 117 for Metals) (Metric).
MIL-B-16541	-	Bronze, Valve: Castings.
MIL-P-16789	-	Packaging of Pumps, Including Prime Movers and Associated Repair Parts.
MIL-M-17060	-	Motors, 60 Hertz, Alternating Current, integral-Horsepower, Shipboard Use.
MIL-L-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-T-17523	-	Turbine, Steam, Auxiliary (and Reduction Gear) Mechanical Drive.
DOD-P-17545	-	Primer Coating, Alkyd-Red Lead Type. Formula No. 116 and Formula No. 116D (Metric).
MIL-B-17931	-	Bearings, Ball, Annular, for Quiet Operation.
MIL-C-20159	-	Copper-Nickel Alloy Castings (UNS NO. C96200 and C96400).
MIL-E-23457	-	Engines, Diesel Marine, Propulsion and Auxiliary, Medium Speed.
MIL-G-24139	-	Grease, Multipurpose Water Resistant.
MIL-P-24377	-	Packing Material, Asbestos, Braided, Impregnated with PTFE (Polytetrafluoroethylene) Surface Lubricated.
MIL-B-24480	-	Bronze, Nickel-Aluminum Castings, for Sea Water Service.
DOD-G-24508	-	Grease, High Performance, Multi-Purpose (Metric).

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MILITARY (Continued)

MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-595 - colors.

MILITARY

DOD-STD-100 - Engineering Drawing Practices.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
DOD-STD-2147 - Technical Repair Standards (TRS); Hull, Mechanical, and Electrical (HM&E), Preparation of (Metric).

HANDBOOK

MILITARY

MIL-HDBK-267 - Guide for Selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment.

2.1.2 Government drawing and publication. The following Government drawing and publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

B-214 - Root Connections for Attaching Piping.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-001-7000 - Fabrication and Inspection and Test of Brazed Piping Systems.

(Copies of specifications, standards, handbook, drawing, and publication required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

HYDRAULIC INSTITUTE
Test Code of Standards - Centrifugal Pump Section.

(Application for copies should be addressed to Hydraulic Institute, 712 Lakewood Center N., 14600 Detroit Avenue, Cleveland, OH 44107.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 2200 Mill Road, Alexandra, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

[Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 General design.

3.2.1 The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. It is the intention of this specification to obtain pumps of such design that they will operate over a long period of years with a minimum of servicing. Where wear or erosion is unavoidable, the parts subjected to such wear or erosion shall be of the best materials available for the purpose in order to reduce these detrimental effects to a minimum.

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3.2.2 Couplings shall be of all metal construction and of non-sparking material if installed in a hazardous area.

3.2.2.1 Flexible couplings shall be keyed to the shafts and secured by lock nuts. For shafts 1-1/2 inches in diameter and larger, the hubs shall be fitted on a taper with keys parallel to the tapers.

3.2.2.2 Rigid couplings shall have fitted bolts and the coupling flanges shall be marked at assembly to insure proper reassembly after overhaul of the units.

3.2.3 Piping and valves. Valves, flanges, and fittings (see 6.2.1) for pipe connections shall conform to MIL-STD-777.

3.2.4 Diesel engines. Diesel engines for driving pumps (see 6.2.1) shall be in accordance with MIL-E-23457.

3.2.5 Turbines.

3.2.5.1 Turbines for driving pumps (see 6.2.1) shall be in accordance with MIL-T-17523.

3.2.6 Motors.

3.2.6.1 Motors for driving pumps shall be of the type and characteristics specified in the contract or order (see 6.2.1), and shall conform to MIL-M-17060, service A, for alternating current motors.

3.2.6.2 Motor rotors for use on dual turbine and motor driven pumps shall be designed and tested for a speed 25 percent in excess of rated turbine speed.

3.2.6.3 The horsepower rating of each motor shall not be less than the maximum brake horsepower of the driven pump under any condition from shut-off to free delivery. The actual motor rating shall be in accordance with Navy standard motor ratings.

3.2.6.4 Motor insulation shall be class B or F as specified (see 6.2.1). The stator windings shall be insulated and varnish treated to provide a sealed insulation system. Motor temperature rise shall not exceed 70 degress Celsius (°C) in a 50°C ambient temperature regardless of the class of insulation specified herein.

3.2.7 Controllers.

3.2.7.1 Motor controllers shall be of the characteristics specified (see 6.2.1) and shall conform to MIL-C-2212 for alternating current controllers.

3.2.8 Shock and vibration.

3.2.8.1 When specified (see 6.2.1), the complete pump unit shall pass the grade A shock test specified in accordance with MIL-S-901.

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3.2.8.2 When specified (see 6.2.1), shock mounts (resilient mounts for shock attenuation) shall be used. When noise attenuation mounts are specified, the pump shall pass the specified shock test solidly mounted.

3.2.8.3 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance. Hole diameters shall be no more than 1/32 inch larger than the bolts for sizes up to and including 3/4 inch, and no more than 1/16 larger than bolts of greater than 3/4 inch size. Mounting bolts for fastening the equipment shall conform to grade 2 or better of MIL-S-1222.

3.2.8.4 Vibraiton. Pumps shall not be damaged or caused to malfunction by either the environmental vibration or by internally excited vibration. Units furnished under this specification shall be tested to meet the requirements in accordance with MIL-STD-167-1, type I and type II (see 6.2.1).

3.2.8.5 Design, construction, and workmanship of the equipment shall be such that noise levels under all conditions of operation will meet the air-borne noise, and structureborne vibration noise requirements in accordance with MIL-STD-740 or to the requirements of the applicable ship detail specification, whichever is lower.

3.2.8.6 Locking devices. Pump internal threaded fasteners and threaded machine elements shall be secured by locking devices (see 6.2.1) where continuous operation under the conditions of shock, vibration, and temperature specified herein depend on maintaining tight connections of parts. Set screws shall be secured by staking.

3.2.9 Threaded parts such as bolts, studs and nuts shall conform to FED-STD-H28.

3.2.10 Welding shall be in accordance with MIL-STD-278

3.2.11 Silver brazed joints shall be in accordance with NAVSEA 0900-LP-001-7000. For joints 1/2 inch and above, preinserted ring fittings shall be used. For sizes below 1/2 inch, face feed fittings may be used.

3.2.12 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.3 Detail requirements.

3.3.1 Type I pumps.

3.3.1.1 Pumps shall be of the double suction volute design and may be either single or two-stage.

3.3.1.2 Pumps shall be for parallel operation and shall have constantly rising head-capacity characteristic curves. The characteristic curve shall be such that at constant rated speed the total head at shutoff will be not less than 10 percent nor more than 20 percent above total head at rated capacity.

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3.3.1.3 Pump operating speed shall be approximately 3,550 revolutions per minute (r/min).

3.3.1.4 Mounting.

3.3. 1.4.1 Pump unit shall be horizontally mounted.

3.3.1.4.2 When specified in the contract or order (see 6.2.1), each pump and separable driving unit, complete with all appurtenances, shall be mounted as a common bedplate.

3.3.1.4.3 Bedplates shall be of cast steel, class CW in accordance with MIL-S-15083, or of structural steel fabricated by welding.

3.3.1.4.4 Bedplates shall be rigid to permit handling shipment, and installation of the units on board ship, without disturbing the alignment of the assembled units; and such that the normal distortion, weaving or vibration of the ship's supporting structures cannot cause misalignment between the pumps and driving units.

3.3.1.4.5 Bearing and seating surfaces of bedplates shall be machine finished.

3.3.1.4.6 Each component part of an assembled unit supported directly by a bedplate shall be doweled thereto to facilitate reassembly and maintenance of alignment.

3.3.1.5 Couplings.

3.3.1.5.1 When no bedplates are specified (see 3.3.1.4.2), pumps shall be furnished with a jackshaft utilizing a rigid coupling on the pump end and a flexible coupling on the driver end.

3.3.1.5.2 Horizontal flexibly coupled units with bedplates shall have coupling guards provided. Guards shall permit ready access to the coupling for lubrication and inspection.

3.3.1.6 Bearings.

3.3.1.6.1 Radial and thrust bearings shall be of the rolling contact type. Rolling contact bearings shall be of the ball type in accordance with FF-B-171. Roller bearings may be recommended if use of ball bearings is impractical. If recommended, roller bearings shall be in accordance with FF-B-187. Rolling contact bearings shall be selected to result in a minimum L-10 life of 10,000 hours calculated in accordance with FF-B-171.

3.3.1.6.2 Rolling contact bearings, specially selected for quiet operation in accordance with MIL-B-17931, shall be used for pumps required to meet special noise requirements (see 6.2.1).

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3.3. 1.6.3 Thrust bearings.

3.3.1.6.3.1 Thrust bearing shall be installed on each pump for counter-acting any unbalanced hydraulic or mechanical thrust in either direction. In this connection consideration shall be given to the fact that the rolling, pitching, or listing of a ship at sea may introduce thrust loads even though the unit is in hydraulic balance.

3.3.1.6.3.2 Thrust bearings shall be of the combined radial and thrust rolling contact type. When pivoted, segmental-type thrust bearings are used, the design shall permit renewal or refitting of shoes without removal of shafts .

3.3.1.6.3.3 Combined radial and thrust bearings shall be type 134, class 1, ball bearings in accordance with FF-B-171.

3.3.1.6.4 Installation.

3.3.1.6.4.1 Bearings shall be installed in housings separate from and independent of the stuffing boxes and with adequate protection from gland leakage. Means shall be provided to prevent escape of oil around the shaft. Baffles, wipers and related parts shall be readily renewable. In vertical pumps, rolling contact bearings, where used, shall be installed external to the pump between the coupling and the gland seal.

3.3.1.6.4.2 Bearing brackets shall be either cast integral with the casing, securely bolted to the casing on a machined shoulder, or held in alignment with the casing by at least two heavy dowels and securely bolted thereto. The use of bolts alone for securing brackets is prohibited.

3.3.1.6.4.3 Bearing housings shall be cast integral with the bracket or secured thereto in such a manner as to insure alignment. Bearing housings to bracket seats shall be machined such that the precision of bearing alignment is insured.

3.3.1.6.4.4 Bearing lubrication and drain connection shall be located so as to provide a path which will force new grease into the bearing and direct old displaced grease out of the drain.

3.3.1.7 Lubrication. Lubrication of rolling contact bearings shall be in accordance with FF-B-171.

3.3. 1.7.1 Rolling contact bearing lubrication method shall prevent bearing failure resulting from over greasing.

3.3.1.7.2 Rolling contact bearings having a DN value in excess of 200,000 shall be oil lubricated unless the contractor can demonstrate to the design review activity that the specific grease lubricated bearing application has been proven in previous Navy shipboard service. Grease lubrication is preferred where DN is less than 200,000.

Where: D = Diameter of bearing bore in millimeters.
N = Revolutions per minute (r/min).

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3.3.1.7.3 Where forced feed lubrication is provided for turbine-driven pumps, the lubricating system shall be in accordance with requirements of the applicable turbine specification (see 3.2.5). Grease lubricants shall be in accordance with DOD-G-24508 or MIL-G-24139. Oil lubricants shall be in accordance with MIL-L-17331. Design shall be such that any qualified grease under DOD-G-24508 or MIL-C-24139, or oil under MIL-L-17331 can be used interchangeably with any other qualified product to the same specification. Bearings shall be relubricated without removing the bearings.

3.3.1.7.4 Thrust bearing housings, particularly for bearings of the pivoted, segmental-type, shall be so designed that the thrust bearings will be adequately lubricated immediately when the shaft starts to turn. This shall be accomplished by properly locating oil supply and drain pipes, and by adequate sealing of the bearing housings so that an adequate supply of oil will remain in the bearing housings.

3.3.1.8 Casings.

3.3.1.8.1 The casings shall be divided on the shaft centerline to facilitate removal of the rotating elements without disturbing suction and discharge connections.

3.3.1.8.2 Sufficient clearance shall be provided around bolt heads and units to permit the use of ordinary tools.

3.3.1.8.3 Fitted bolts on heavy dowel pins shall be provided to insure maintenance of alignment of casing halves in reassembly.

3.3.1.8.4 Forcing bolts shall be provided for breaking joints.

3.3.1.8.5 Pumps shall be fitted with connections to permit complete draining of the pumps.

3.3.1.8.6 Pumps shall be fitted with vent connections to prevent entrapment of air or vapor which would adversely effect the operation of the pump. Special attention should be paid to vent location so that the seal lubrication requirements of 3.3.1.11.4 are met during pump start up and operation.

3.3.1.8.7 Casings shall be fitted with removable wearing rings.

3.3.1.8.8 Pumps shall be fitted with flanged connections conforming to Drawing B-214 for piping up to suction and discharge pressure gauges (see 6.2.1).

3.3.1.9 Suction and discharge connections.

3.3.1.9.1 Suction and discharge connections shall be flanged for mating with flanges in accordance with MIL-STD-777 and as specified in 6.2.1.

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3.3.1.9.2 Suction and discharge connections shall be located in the bottom or fixed half of casings.

3.3.1.10 Impellers and shafts.

3.3.1.10.1 Outside surfaces of impellers shall be smooth finished, and inside surfaces shall be file finished insofar as practical.

3.3.1.10.2 Impellers shall be keyed on the shaft and securely held against lateral movement by lock nuts or other means approved by the design review activity.

3.3.1.10.3 Impellers shall not be furnished with wearing rings. Impeller hub wearing surfaces shall have material thickness to permit reducing the diameter of the impeller hubs by as much as 0.050 inch to accommodate undersize casing wearing rings to restore design running clearance.

3.3.1.10.4 Shaft threading, except threading for rolling contact bearing locknuts, shall be counter to the direction of rotation. Shaft threading for rolling contact bearing locknuts may be clockwise or counterclockwise.

3.3.1.10.5 Shafts shall be fitted with shaft sleeves in way of stuffing boxes and diaphragm bushings.

3.3.1.10.5.1 Shaft sleeves shall be securely locked at the shaft and sealed to prevent leakage between the shaft and sleeves.

3.3.1.11 Stuffing boxes and mechanical shaft seals.

3.3.1.11.1 Pumps shall be provided with mechanical shaft seals, magnetic type seals shall not be used. Each seal shall be provided with a solid gland plate designed to house two or more rings of emergency packing for use in the event of mechanical seal failure. The packing shall be in accordance with MIL-P-24377. Gland design shall be such that the packing rings may be inserted without removing the mechanical shaft seal. The mechanical shaft seal shall have tungsten carbide against a carbon sealing surface.

3.3.1.11.2 Space shall be provided between bearings and stuffing boxes to permit easy examination of mechanical seals and bearings.

3.3.1.11.3 Pumps shall be configured so that normal gland leakage, whether from mechanical seal or when operating under emergency packing, shall be collected and piped to waste.

3.3.1.11.4 For those installations where the mechanical seal is in contact with the impeller, the mechanical seal internal parts must be of the same material as the impeller.

3.3.1.11.5 Mechanical shaft seals shall be positioned on the shaft by means of stub or step sleeves. Mechanical shaft seals shall not be positioned by use of setscrews alone.

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3.3.1.11.6 The emergency packing gland shall be set up by nuts threaded on studs.

3.3.1.11.7 The design shall insure that positive liquid pressure is supplied to the seal faces under all conditions of operation and that there is adequate circulation of the liquid at the seal faces to minimize the deposit of foreign matter on the seal parts. The pump certification data shall contain certification by the seal manufacturer that there will be adequate circulation of liquid and adequate lubrication at the seal faces when the seal is installed as shown on the drawing.

3.3.1.12 Materials. Materials of construction shall be as specified in table I.

TABLE I. Materials of construction.

Part	Material	Specification
Casing	Copper nickel alloy (70-30)	MIL-C-20159
	Nickel-aluminum-bronze	MIL-B-24480
Impellers	Nickel-copper-alloy	QQ-N-288, comp E
Shaft	Nickel-copper-alloy	QQ-N-281
Shaft sleeves	Nickel-copper-aluminum-alloy 265 or higher Brinell	QQ-N-286
Casing wearing rings	Nickel-copper-alloy	QQ-N-288, comp B or D
Glands and throat bushings	Nickel-copper-alloy castings	QQ-N-288, comp E
Bearing housing	Copper-nickel alloy (70-30)	MIL-C-20159
Mechanical seals	Nickel-copper-alloy	QQ-N-288 or QQ-N-281

3.3.2 Type II pumps.

3.3.2.1 Pumps shall be of the multi-stage deep well turbine type.

3.3.2.2 Pumps shall have constantly rising head-capacity characteristic curves, The characteristic curve shall be such that at constant rated speed the total head at shut-off will be not less than 10 percent above total head at rated capacity.

3.3.2.3 When a pump of the self-priming type is specified, a special first stage priming impeller and an air separation chamber between the first and second stages should be provided. The first stage impeller shall function as a vacuum pump, with a maximum suction lift of 20 feet of liquid, for removing air or vapor from the suction well and suction piping during initial priming operation. The separation chamber shall be furnished with a suitably sized pipe to be used for venting back to the tank.

3.3.2.4 Mounting.

3.3.2.4.1 Pump units shall be vertical mounted by means of a flange bolted to the deck. This flange shall be cast integral with the pump discharge assembly.

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3.3.2.4.2 Pumps shall be either connected through a flexible coupling to a steam turbine or directly connected to an electric driving motor through a hollow motor shaft. If a right angle gear drive is utilized, a hollow shaft shall connect the gear drive to the prime mover. Pumps connected to the prime mover by hollow shafts shall be furnished with an adjusting nut on top of the pump shaft to facilitate the positioning of the pump impeller.

3.3.2.4.2.1 When pumps are driven by an electric motor, a nonreverse ratchet shall be provided to prevent reverse rotation of the pump.

3.3.2.5 Pump discharge head assembly.

3.3.2.5.1 The discharge head assembly shall be for mating with the motor or gear flange and shall be equipped with a separate base flange to permit removal of the pump head without disturbing the deck seal.

3.3.2.5.2 The discharge connection shall be flanged in accordance with MIL-STD-777.

3.3.2.5.3 Stuffing boxes and mechanical shaft seals.

3.3.2.5.3.1 Stuffing boxes and mechanical seals shall be in accordance with 3.3.1.11.

3.3.2.6 Column assembly.

3.3.2.6.1 The column assembly shall be suspended from the head assembly and shall support the bearing spiders.

3.3.2.6.2 Columns shall be constructed of uniform lengths of pipe not exceeding 5 feet unless otherwise specified in the contract or order (see 6.2.1). Column sections shall be flanged at both ends.

3.3.2.7 Shaft assembly.

3.3.2.7.1 Shafts shall be furnished in 5 foot sections unless otherwise specified, (see 6.2.1) and held together by means of threaded rigid couplings.

3.3.2.7.2 Each section of shaft shall be steadied by a sleeve type bearing.

3.3.2.7.3 Bearings shall be held in place by the use of a bearing spider.

3.3.2.7.4 Bearings shall be lubricated by the liquid pumped.

3.3.2.8 Bowl assembly.

3.3.2.8.1 Bowls shall be of the single stage type and interchangeable with each other.

3.3.2.8.2 Bowls shall be flanged and bolted together.

3.3.2.8.3 Suction bowls shall be flared and flanged to the lower bowl assembly.

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3.3.2.9 Impellers. Impellers shall be of the enclosed type and held to the shaft by means of split taper-locking bushings.

3.3.2.10 Materials. Materials of construction shall be as specified in table II.

TABLE II. Materials of construction.

Part	Material	Specification
Column	Steel-galvanized	WW-P-404
Bowl	Valve bronze	MIL-B-16541
	Gun metal	QQ-C-390, grade 903
Impeller	Nickel-copper-alloy	QQ-N-288, comp E
Shaft	Nickel-copper-alloy	QQ-N-281
	Nickel-copper-aluminum alloy	QQ-N-286

3.3.2.10.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Painting.

3.4.1 External unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment, in accordance with DOD-P-15328, and one coat of zinc chromate primer, TT-P-645, or one coat of red lead primer, DOD-P-17545, followed by a finish coat of light gray equipment enamel, MIL-E-15090. Color shall conform to number 26307 of FED-STD-595.

3.4.2 Painting of external surfaces of nonferrous parts of pumps shall not be required.

3.5 Identification plates.

3.5.1 Identification plates shall be furnished on each pump and shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that identification plates of plastic or aluminum will not be acceptable.

3.5.2 Identification plates shall be secured to equipment with corrosion resistant metallic screws or rivets.

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3.5.3 Pump identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Service application.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
    (1) Capacity in gallons per minute.
    (2) Total head in pounds per square inch (lb/in^2).
    (3) Suction pressure.
    (4) Specific gravity or viscosity of fluid pumped in saybolt seconds universal.
    (5) Speed of shaft in r/min.
    (6) Brake horsepower.
    (7) Hydrostatic test pressure.
    (8) Impeller diameter.
(f) Contract number (and item number for multiple unit orders).
(g) National stock number.
(h) Section for inspector's stamp.
(i) NAVSEA technical manual number.

3.5.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.6 Tools which are, available in the Federal Supply Catalog, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material. The number of sets of special tools of the pump units shall be as specified (see 6.2.1). Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of the catalog may be consulted in the office of the DCASMA). Special tools shall include the following, if applicable:

(a) Complete set of special wrenches of forged steel with hardened jaws.
(b) One impeller pulling tool.
(c) One set of babbitting mandrels with collars, complete for renewing white metal of all bearings.
(d) One set of ball bearing pulling tools.
(e) Any other special tools necessary for the proper maintenance of the units in service.

3.7 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.8 through 3.9.9.

3.8 Drawings. When specified (see 6.2.1) the contractor shall prepare drawings in accordance with the data ordering document included in the contract or order (see 6.2.2) and the unique features specified in 3.8.1 through 3.8.4 shall be included.

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3.8.1 Drawing content.

3.8.1.1 General requirements.

3.8. 1.1.1 "Manufacturer's use only" notes. Information intended for manufacturer's use only shall be so designated, Conversely, on a commercial drawing with wide usage it is permissible to designate portions thereon, "For Navy use only."

3.8.1.2 Design drawings. Design drawings shall consist of the drawings described in 3.8.1.2.1 through 3.8.1.2.4.

3.8.1.2.1 Sectional assembly drawings. Sectional assembly drawings shall include a sectional assembly with references to the list of material for identification of individual parts. All running clearances shall be shown for reference. (The radial clearance equals one half the diametral clearance.) This may necessitate some side or partial view in order to show parts not otherwise shown in the main section view. Tightening torques and thread lubrication requirements for threaded fastners shall be shown on the assembly drawing.

3.8.1.2.2 List of materials. The list of material shall include every part required in the pump assembly, including those parts not required to be detailed. The list of material shall include an indication of each part required by this specification to be furnished as an on board repair part (this is not a repair parts list). The list of material shall include the manufacturer's drawing number and service part number for all parts for which detail drawings are required (see 3.8.1.2.3) as well as quantity required, material type, and material specification for all parts. Parts for which detail drawings are not required shall be identified as to dimensions, The list of material may be shown on the sectional assembly drawing.

3.8.1.2.3 Detail drawings. Detail drawings shall be furnished of all parts and sub-assemblies necessary for evaluation of the equipment, and of all parts necessary for maintenance, and overhaul of the pumps. Details of these parts shall be so complete as to permit emergency manufacture by a Naval shipyard without assistance from the original manufacturer. Details of pump casing shall be furnished, but the dimensioned development of cored hydraulic passages need not be shown. Sub-assemblies whose parts cannot be acquired or serviced individually, should be shown as a single part and so indicated, Multi-detail drawings are preferred, but mono-detail drawings may be used.

3.8.1.2.3.1 Impeller machining and inspection drawing. Detail drawings shall explicitly require that each impeller, including spare and replacement impellers, be dynamically balanced to less than the maximum ounce-inch residual imbalance allowed in accordance with MIL-STD-167-1 (type II) for the impeller's weight and rotational speed.

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3.8.1.2.3.2 Drawings showing the application of studs requiring the use of a locking resin shall have a notation requiring: (1) manufacturer's recommendation regarding the use of primer, (2) manufacturer's recommended primer, and (3) the minimum and maximum waiting times associated with the use of the primer and resin. The notation shall require strict adherence to these requirements.

3.8.1.2.4 Outline drawings and certification data.

3.8.1.2.4.1 Outline drawings and certification data shall be in the form of supplemental drawings to all sectional assemblies and detail. Separate outline drawings, drawing lists, and certification data shall be furnished under each contract or order unless the complete equipment covered by the outline drawing and the referenced drawings are in fact identical in all respects.

3.8.1.2.4.2 The outline drawings in addition to the certification data required in accordance with DOD-D-1000 and DOD-STD-100 shall contain the following:

(a) Dimensional outline assembly drawing of the pump with its prime mover, bedplate and attached auxiliaries.
(b) Complete performance data of pump, prime mover, and attached auxiliaries, if applicable.
(c) Complete equipment performance curves, based on actual tests (the original submission may show design performance curves vice test curves).
(d) Weights of major sub-assemblies over 50 pounds, for example, rotor, upper half-casing, impeller and the weight of the complete unit including driver, dry and in operating condition,
(e) Shipbuilder's connections showing size, type and dimensions of flanges.
(f) Center of gravity of the pump component and of the complete assembly.
(g) Radii of gyration of complete assembly about each of the three principal axes.
(h) Identification of system in which installed.
(i) Critical speed.
(j) Maximum thrust loads.
(k) Endurance tests, high-impact shock tests and non-destructive tests (if performed) and the design review agency letters approving these reports or extension of any tests.
(l) Overall dimensions of the complete unit including locating of all fittings and connections, the space required for removal and replacement of parts for maintenance, and mounting information.
(m) The certification by the mechanical seal manufacturer required (see 3.3.1.11.7).

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3.8.1.2.5 On board repair parts list. A list of on board repair parts shall accompany the outline drawings. This list shall be prepared on provisioning list forms in accordance with MIL-P-15137. This list shall be furnished whether or not the parts are required in the pump contract or order. This list shall not be modified to indicate parts or quantities of parts furnished under supplementary or separate contracts or orders. The list is intended as a record of recommendations of the manufacturer and purchaser at the time of equipment drawing approval. The list of on board repair parts actually purchased shall be prepared as required in accordance with MIL-P-15137.

3.8.1.3 Basic design drawings. Basic design drawings covering pumps under this specification are intended for submittal to the design review agency prior to and independent of invitations for bids. As such these are different from design drawings as described in 3.8.1.2. Each drawing may be a range of sizes of pumps so long as they are of the same basic design.

3.8.1.3.1 Contents of basic design drawings. These drawings shall consist of small scale (not necessarily to scale) plan and elevation views showing over-all dimensions. If the plans cover a family of pump sizes, all of the same design, the dimensions may be tabulated. The range of capacities, pressure and speeds shall be shown. The main part of this drawing shall be an undimensioned sectional assembly with a complete list of material which shall include every part required in the pump assembly. If materials of some parts may change with various pump characteristics or with differing fluids to be pumped, a table shall be included to show these variations, Alternate sub-assembly arrangements of different parts may be included as desired.

3.8.1.3.2 Detail drawings of individual parts are not required in connection with basic design drawings, but may be included if desired by the manufacturer for clarification, or to obtain advance approval of the design agency.

3.8.1.4 The delivered drawings shall identify and certify the following:

(a) Whether or not there is an adhesive or other setting compound factory-installed on the elastomer bellows or O-ring that seals between the mechanical seal and the impeller hub or shaft sleeve.
(b) The description of any such adhesive or setting compound.
(c) Any time limit associated with that compound when installed on a mechanical seal in stowage and any time limit between initial wetting of that compound and the final positioning of the mechanical seal in the pump.
(d) The type and amount of lubricant, if any, to be used to facilitate axial positioning of the mechanical seal in the pump during seal installation. The lubricant shall be one of the standard lubricants or compounds as specified in MIL-HDBK-267. The drawing shall require the manufacturer to use that lubricant during pump assembly.

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3.8.2 Material identification,

3.8.2.1 Preferred material reference. Where materials of identical or equal quality can be more than one specification or standard, the drawings need reference only one such specification or standard. In selecting the specification or standard to be referenced the following is the order of preference:

(a) Federal specification or standard.
(b) Military specification or standard.
(c) Industry and technical society specification or standard.
(d) Manufacturer specification or standard.

3.8.2.2 Material substitutions. Where materials other than covered by (a), (b), and (c) of 3.8.2.1 are referenced and approved by the design review agency, the drawings shall show the complete chemical and physical properties of the approved material. In addition, the drawings shall identify the material in terms of the nearest Federal or Military specification, in order to enable Naval repair facilities to make emergency repairs as necessary from Navy material stocks.

3.8.3 Drawing identification. Prime vendors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in all cases where the parts delineated thereon are produced by the subcontractor. The prime vendor shall not add his drawing number to the drawing except as an unofficial reference outside the drawing border or margin.

3.8.3.1 Prime vendors who purchase semi-finished parts from subcontractors for final production, test or selection phases in their own shops, have the option of using as the single drawing identification either their own title block and drawing or the title block and drawing number of the subcontractor, but not both.

3.8.4 Drawings of drivers and associated equipment shall be in accordance with the specifications covering those equipments.

3.9 Technical manuals.

3.9.1 Unless otherwise specified in the contract or order (see 6.2.1), technical manuals shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2). The modifications and clarifications specified in 3.9.2 through 3.9.5 shall apply. A separate manual shall be prepared for each different pump unit except in special cases when manuals covering more than one pump may be approved. When specified in the contract or order (see 6.2.1), manuals shall include pump, driver, controls and all appurtenances and all applicable drawings. Performance curves for final manuals may be furnished as insert pages after delivery of hardware.

3.9.2 Manuals covering pumps driven by turbines or engines shall include drawings and full operation and maintenance instructions covering the turbines, engines, governors and any other applicable accessories.

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3.9.3 Manuals covering pumps driven by conventional electric motors with conventional controllers need not include operation and maintenance instructions covering the motors and controllers. They shall, however, include master drawings and certification data covering the motors and controllers . Complete operation and maintenance instructions covering electrical equipment shall be included when specifically required (see 6.2.1).

3.9.4 Each manual shall include not less than the following illustrations covering the pump and driver:

(a) Sectional assembly drawing.
(b) Outline drawing.
(c) Complete list of material corresponding to the sectional assembly drawing.
(d) Certification data.
(e) A minimum of two photographs of the complete unit, pump and driver, taken 180 degrees apart on a horizontal plane. Where several sizes of identical design units are being supplied, photographs of one design unit only need be supplied.

3.9.5 The alignment procedure instructions in the manual shall describe in complete detail the means by which the required alignment clearance at sleeve bearings and wearing rings is to be established. The manual shall prescribe the maximum permissible clearances which can be tolerated in the aforementioned areas with the pump in the assembled condition. Exceptions to any part of the aforementioned requirements in any specific area may be had only by providing a written technical justification for said exception to the design review agency for approval. The manual shall specify the maximum allowable clearance for wearing rings and sleeve bearings before replacement of the parts. In addition, the manuals shall contain:

(a) Lubrication schedule including lubricant and procedure.
(b) A statement directing the vendor to see the Centrifugal Pump Chapter of the Naval Ships Technical Manual for the criteria for replacing mechanical seals due to leakage.
(c) The maximum allowable mechanical seal leakage rating shall be specified.
(d) Torque values and sequence required to maintain proper seal and alignment (for type I pumps).
(e) In-field dynamic balancing instructions where provisions for balancing in field are required.
(f) The mechanical seal installation procedure including complete information regarding adhesive, the adhesive time limits and lubricant (see 3.8.1.4), and any additional guidance needed to avoid causing an excessive preload on the mechanical seal.

3.9.6 Technical repair standard. A technical repair standard (TRS) shall be prepared upon initial introduction of a new equipment or introduction of an equipment that has an approved TRS if major or minor configuration changes have been approved that would effect the adequacy of the standard for use in new equipment repairs (see 6.2.2).

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3.9.7 TRS technical content shall include sufficient technical details to enable a repair, maintenance, or overhaul activity to restore the equipment dimensions, clearances, and tolerances such that the equipment is capable of performing its function as originally specified and is capable of being logistically supported by the DoD logistics support system.

3.9.8 The TRS format and content shall conform to DOD-STD-2147 and shall be approved and validated in accordance with this standard.

3.10 Workmanship. The equipment, including all parts and accessories, shall be manufactured and finished with first class workmanship in all respects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all Inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5, The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 The inspection requirements shall consist of the following:

(a) Completed parts examination (see 4.1.3.1).
(b) First article inspection (see 4.3).
(c) Tests (see 4.6).
    (1) Shop tests. To be conducted on each pump supplied under the contract or order (see 4.6.1).
    (2) Performance evaluation tests. To be conducted on one pump of each type, design and size on each contract or order (see 4,6.5).
    (3) Design evaluation tests. To be conducted on one pump of each type, design and size (see 4.6.6).

4.1.3 Inspection system. The contractor shall prepare an inspection system in accordance with the data ordering documents included in the contract or order (see 6.2.2).

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4.1.3.1 The inspection system shall include a completed part examination procedure whereby each functionally and dimensionally critical part and assembly, including the final product, is examined on completion in accordance with a completed part examination check list accompanying each part or assembly (see 4.5).

4.1.4 Variations. No variations or deviations from approved design drawings as specified in 3.8.1.2 will be permitted.

4.1.5 Mercury-prohibition. Pumps, drivers, and auxiliary parts furnished by the pump contractor shall be free of mercury contamination and free of functional mercury (see 6.2.1). During the manufacturing processes, checks, examinations and tests, the product being offered for acceptance shall not come in direct contact with mercury, any of its compounds, nor with any mercury containing device, such as gauges and thermometers. The manufacturer shall certify that the product, when shipped, is free from mercury contamination.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the following test specifications:

(a) Examination test as specified in 4.5.
(b) Performance evaluation tests as specified in 4.6.5 through 4.6.5.3.
(c) Design evaluation tests as specified in 4.6.6 through 4.6.6.5.

4.3.1 First article test report. The contractor shall prepare a first article test reports in accordance with the data ordering document (see 6.2.2),

4.4 Quality conformance inspection. Each pump (and subcomponents of each pump, if applicable) shall be examined as specified in 4.5 and tested as specified in 4.6.1 through 4.6.4. Quality conformance inspection shall be fully documented by a prescribed test procedure and a written record of the performance and findings of each test (see 6.2.2). A compilation of the documentation of all the quality conformance tests performed on each pump shall be furnished in duplicate (see 6.2.2).

4.5 Examination. Pumps shall be examined in accordance with a completed part examination check list accompanying each part or assembly. The examination shall be performed by inspection personnel not engaged in the fabrication and assembly of the part. The check list for each part shall include accurate part identification and shall list the specific attributes which determine the part's acceptability. The check list shall be arranged in tabular form, calling forth separately the following examination assignment, each accompanied by the specific attributes of acceptability which shall be checked or measured and the findings recorded:

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(a) Dimensions. They shall be as indicated on the design drawing. Rotating parts shall be examined for balance and maximum permissible run-out as indicated on the drawings.
(b) Materials. Visual examination for compliance with design drawings. Any appearance of nonconformance shall be further verified by appropriate tests.
(c) Cleanliness. Examination for cleanliness shall apply equally to parts examination prior to assembly as well as subassemblies and the completed product.
(d) Missing parts. This examination applies primarily to the completed product prior to shipment.
(e) Assembly. During assembly, positioning tolerances and wear ring/impeller running clearances shall be verified.
(f) Alignment. The equipment and its subassemblies shall be examined to insure that the alignment is as specified.
(g) Adjustment. The adjustment of safety, control, and monitoring devices shall be verified and the settings recorded. Settings which should not be tampered with, once adjusted, shall be tagged with an appropriate precautionary warning.
(h) Preservation. The preservation used, the method of application, and recommended procedure for removal of preservative prior to placing the unit in service shall be stated on a supplemental sheet and attached to the check list.

4.6 Tests. The contracting agency and Government representatives shall have the right to examine the facilities at the contractor's plants and at the contractor's subcontractor's plants, and to witness all tests specified herein. Failures, deficiencies, and discrepancies revealed during the performance of the specified tests and the corrective measures taken, shall be recorded and fully documented in the applicable test records and test reports (see 6.2.2). After correction of deficiencies, tests shall be repeated to the extent necessary to ascertain acceptability for the modified pump. Major failures indicative of the design deficiency (as distinguished from shop error or faulty workmanship) shall be reported to the design review agency before a correction is made. Except where specifically required herein to be tested with the actual device, tests of turbine driven pumps may be conducted using a substitute driver provided the actual device separately passes the tests required by its specification.

4.6.1 Hydrostatic pressure test. All pressure boundary parts shall be tested hydrostatically to a pressure one and one-half times the maximum discharge pressure at maximum submergence, but in no case less than 50 lb/in^2. The hydrostatic test pressure shall be maintained for at least 30 minutes or longer as necessary for examination of entire casing.

Acceptance criteria: The pump shall exhibit no leakage through the casing material or joints. The leaking rate through a mechanical seal shall not exceed 5 drops per minute.

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4.6.2 Over-speed test. Each pump driven by a variable speed driver capable of over-speeding shall be operated continuously for 30 minutes at a speed 25 percent above maximum design operating speed. The pump need not be under load except as necessary to prevent damage or injury. Continuous shaft and rigid-coupled pumps shall be tested with the driver. Pumps separately driven through flexible couplings and geared units may be tested by separate driver, gear unit, and driver over-speed tests. The dynamic balance shall be checked by the use of a vibration indicator of a type whereby vibrations of an amplitude of 0.001 inch may be readily observed on a scale.

Acceptance criteria: Neither the pump nor the driver shall exhibit abnormal noises or roughness of operation. Maximum acceptance readings of the vibration indicator shall not exceed the limits specified in accordance with type II of MIL-STD-167-1 for this equipment.

4.6.3 Mechanical soundness and capacity test. This test shall be conducted, recorded, and reported in accordance with the centrifugal pump Rating Standard and Test Code of the Hydraulic Institute, to the extent that these standards are applicable and are not in conflict with the requirements specified herein. The test record for each pump shall identify the major pump components (pump, gear assembly, driver) by manufacturer's serial number, indicate the diameter of the pump impeller being tested in the pump, include a description (by legend and sketch) of the test stand arrangement including an identifying list of test instruments used and the date when last calibrated, and include the data sheets of all recorded data and sample calculations of data conversion to specified conditions. Test data used for plotting the head-capacity curve shall be corrected to the specified operating conditions, The test shall be performed as follows:

(a) Operate the pump and its driver, if motor driven, continuously at the maximum rated speed and capacity, with the pumped fluid at ambient temperature until bearing temperatures stabilize. Stabilization is defined as three consecutively recorded readings taken over intervals of at least 15 minutes that fall within 3 degrees Fahrenheit (°F) band when adjusted for ambient. The three consecutive readings shall not be constantly rising, The pump operation shall be monitored for proper functioning of safety devices, bearing lubrication, and for smooth running. Vibration shall be measured on the pump shaft.

Acceptance criteria: Unit operation shall be without heating and shall be free of abnormal vibrations and noises. Adequate oil temperature rise in force-feed lubricated bearings shall not exceed 50°F with inlet cooling water to the oil cooler at 85°F. There shall be no abnormal leakage of water or oil. Pump shaft vibration shall meet the acceptance criteria of 4.6.2.

(b) Operate the pump at the maximum rated speed with the pumped fluid at maximum normal temperature from recirculation flow as close to free delivery as practicable and with the minimum specified suction pressure prevailing. The unit shall be

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operated at seven or more test points throughout the full operating range to establish accurately the head-capacity curve at maximum rated speed. The unit shall be operated at each test point until the test values being measured stabilize. In addition to the above testing, determine the required net positive suction head at a maximum rated speed at maximum rated flowrate.

Acceptance criteria: In no case shall the pump deliver less than the specified requirements. The total head at all capacities on the curve shall not deviate by more than plus 5 percent or minus 2 percent of rated heads from the corresponding head on the head-capacity curve at maximum rated speed established in the performance test specified in 6.2.1.

(c) Operate the electric motor-driven unit for a minimum of 1 minute in reverse rotation at maximum rated speed.

Acceptance criteria: The unit shall not be damaged by the reverse rotation test. Conformance to hydraulic and noise requirements shall be demonstrated subsequent to the reverse rotation test.

4.6.4 Noise tests. Airborne and structureborne noise tests when specified (see 6.2.1) shall be conducted and reported in accordance with MIL-STD-740. Noise test details, instrumentation, and testing techniques identified in MIL-STD-740 shall be submitted to the contracting activity prior to testing for approval. Noise tests shall be performed with the driver furnished with pump, and tests shall be conducted on all units. Airborne noise tests shall be conducted on the lead unit only on each contract or order.

Acceptance criteria: The unit shall meet the noise level limits specified in 6.2.1.

4.6.5 Performance evaluation tests. One pump of each type, design, and size on each contract or order shall successfully undergo the performance evaluation tests to establish the complete pump performance map and to ascertain compliance with the specified performance requirements. Motor driven units shall be tested with the motor to be furnished with the pump. A substitute driver may be used for turbine driven pumps. When pumps of identical type, design, and size are being produced under two or more contracts within a 2 year period, only one series of performance evaluation tests shall be required for that production run, The following tests shall comprise the performance evaluation tests:

(a) Performance test (see 4.6.5.1).
(b) Mechanical shaft seal and pump alignment procedure test (see 4.6.5.2).

4.6.5.1 Performance test. The performance tests shall be conducted, recorded, and reported in accordance with the requirements of the tests specified in 4.6.3, except that, in addition, a full performance map shall be established. Test data shall be converted to specified operating conditions for plotting of all

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performance curves. The performance curves shall be determined at maximum and minimum operating speed for variable speed pumps and at each operating speed for multi-speed pumps. A full net positive suction head (NPSH) curve is required on the lead production unit and on subsequent units the NPSH shall be determined at design rated capacity only. The following curves shall be established:

(a) Capacity versus total head.
(b) Capacity versus pump efficiency.
(c) Capacity versus brake horsepower.
(d) Capacity versus net positive suction head required.

Acceptance criteria: The acceptance criteria specified in 4.6.3(a) shall be met. The performance map shall exhibit the specified pump performance characteristics. The net positive suction head required shall not exceed the minimum suction head available as specified (see 6.2.1). Controls and safety devices shall function reliably as intended throughout the full operating ranges of capacity and speed.

4.6.5.2 Mechanical seal and alignment test.

(a) The mechanical seal shall be installed and aligned precisely in accordance with the procedures stated in the proposed pump technical manual. Packing shall not be installed during this test. The amount of force required to axially position the mechanical seal onto the impeller hub, the shaft sleeve or whatever it rests on in the assembled condition shall be measured and recorded. If the mechanical seal includes an adhesive or setting compound on the elastomer that seals between the mechanical seal and the impeller hub or shaft sleeve, and if there is a maximum allowed time between initial wetting of that compound and final positioning of the mechanical seal in the pump, then do not perform that final positioning until the maximum allowed time has elapsed.
(b) After installation, the mechanical seal shall be running and the pump operated for at least 4 hours at maximum rated speed with the pumped fluid at maximum operating temperature and with pump suction pressure as high as feasible without requiring pump test loop modifications. The seal leak rate shall then be measured and recorded. Also, the dynamic balance shall be measured and recorded using a vibration indicator of the neon tube or similar type capable of measuring vibrations of 0.001 inch or less. During subsequent pump disassembly, the final axial position of the mechanical seal shall be measured and recorded if the pump design permits.

Acceptance criteria. Mechanical seal leakage shall not exceed 5 drops per minute. The pump and drive shall exhibit no abnormal noises or roughness of operation. Vibration shall not exceed the limits specified in accordance with MIL-STD-167-1, type II, measured on the bearing caps. The mechanical seal shall have been positioned at the axial position recommended by the mechanical seal vendor to generate the proper amount of mechanical seal spring

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force. The mechanical seal vendor shall provide in writing his recommendation regarding the axial position in his concurrence that the mechanical seal alignment test results demonstrated proper positioning of the seal. If it was not practicable to measure the axial position of the mechanical seal, then the seal vendor shall concur that the recorded force used to position the seal during assembly use accepted and shall state the technical basis for considering that force acceptable. A copy of these mechanical seal vendor documents shall be provided as part of the Lest report.

4.6.5.3 Test reports for the performance evaluation tests shall be compiled into a single document. The document shall identify the pump by manufacturer's serial numbers of pump and driver, the contract or order number under which the tests were performed, the ship or class of ships for which the pump is intended if known, and the letter and date of test procedure approval. The test compilation document shall be distributed as specified (see 6.2.2).

4.6.6 Design evaluation tests. One pump of each type, design, and size complete with the driver and all appurtenances and controls shall successfully undergo the design evaluation tests specified herein. Design changes which in the opinion of the design review agency may detrimentally affect the reliability of a previously tested and accepted pump design shall be cause to require new design evaluation tests in part or in full. It shall be at the discretion of the design review agency to require new design evaluation tests when an accumulation of several design changes, each of itself apparently not significant to pump reliability, collectively considered render suspect the validity of any one or all or the previously performed design evaluation tests. An endurance test need not be performed provided that the vendor can demonstrate to the satisfaction of the design review agency concerned that the pump to be offered has been proven in previous Navy shipboard service under conditions equally as strenuous as those specified in the ordering data. The proposed design evaluation test procedures shall be submitted to the contracting agency prior to performing the tests. The design evaluation tests shall consist of the following:

(a) Endurance test (see 4.6.6.1).
(b) Shock test (see 4.6.6.2).
(c) Inclined operation test (see 4.6.6.3).
(d) Vibration tests (see 4.6.6.4).

4.6.6.1 Endurance tests. The pump shall be operated for a period of not less than 100 hours of actual running time with a minimum of 25 starts to ascertain reliability of performance and operation.

4.6.6.1.1 Prior to commencement of the endurance test and immediately after completion of the 100-hour operating run, the pump shall be disassembled to the extent necessary and the critical dimensions and running clearance of parts subject to wear, erosion and derangement, measured, calculated and recorded. Components such as pump impellers and casings subject to erosion, corrosion, cavitation, and wear, the effects of which are not subject to routine measurement, shall be listed in the inspection record and after completion of the test the condition of each component determined and recorded.

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4.6.6.1.2 During the initial and final hours of the endurance test run, noise and performance tests in accordance with 4.6.4 and 4.6.5.1 shall be performed to determine the changes in the pump's performance characteristics and noise signature. Vibration measurements shall be taken at the bearing caps or housings of the pump and driver at equal speeds during the initial and final capacity test to determine the changes in mechanical operation.

4.6.6. 1.3 The endurance test shall not be continuous but shall be interrupted by at least three rest periods of a minimum of 8 hours each. A minimum of 25 starts at full line voltage shall be performed during the course of the test. The remainder of the endurance tests shall be run at maximum rated speed and within plus 20°F, minus 0°F of maximum specified liquid temperature. The pump shall be operated at one-third, two-thirds, and rated capacity in approximately equal time intervals. Operations at rated capacity shall be at minimum specified net positive suction head available, or maximum specified suction lift or vacuum, as applicable.

4.6.6.1.4 The pump shall be monitored during the endurance test to accurately record the conditions of operation, the capacity delivered, the total head developed, the speed at which operated, and the general performance observed. Data shall be collected and the pump inspected at least twice per day of operation. For each periodic inspection, in addition to all measured data, the record shall indicate the following:

(a) Seal leakage rate and the amount of gland leakage collected and drained by the collection area.
(b) The conditions of the bearings (by noise; by feel; and by bearing temperature via probe if the design includes provisions for a probe, otherwise via a surface pyrometer on a normally exposed surface; no disassembly required).
(c) The airborne noise level (normal-abnormal).
(d) The vibration level (normal-abnormal).
(e) The smoothness of operation (normal-abnormal).
(f) All other abnormal findings.
(g) All adjustments made.
(h) Changes made in the conditions or method of operation.

4.6.6.1.5 Endurance test acceptance criteria.

(a) Head-capacity curve at maximum rated speed after 100 hours of pump operation shall conform to the specification requirements and shall show no abnormal deviations from the curve before the 100-hour test.
(b) Unit performance and operation after 100 hours of operation shall be unchanged and normal and meet all specification requirements .
(c) Unit operation at the end of the endurance test shall be smooth and shall exhibit noise and vibration levels that are normal and in conformance with the specification.
(d) Lubrication shall have remained satisfactory throughout the test period. Bearing temperatures shall have remained normal and shall be consistent with their respective bearing clearances and oil and grease limitations.

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(e) Leakage rate from a mechanical shaft seal shall not exceed 5 drops per minute.
(f) Running clearances shall be normal.
(g) Components subject to attack from corrosion, erosion cavitation, and so forth shall be in a condition commensurate with 100 hours of service.
(h) Wear rates for wearing parts, critical for proper operation, shall show a rate of wear for the test period that shall be consistent with the specified design life requirements.
(i) No failures shall have occurred throughout the 100-hour test.

4.6.6.1.6 Post endurance test procedures. The unit subjected to the 100-hour endurance test shall be restored to the as-new condition by replacement of all parts worn beyond the as-new design tolerances. The restored unit shall successfully pass the tests specified in 4.6.1, 4.6.3(a), and 4.6.4, if applicable . The quality conformance test documentation shall indicate that the unit was subjected to the endurance test and subsequently restored, tested, and that it shall be certified as fully conforming to the specification for unrestricted service.

4.6.6.2 Shock test. The pump shall undergo a shock test to ascertain that the pump has the necessary shock resistance. The shock test shall be performed in accordance with MIL-S-901, grade A, and the specific shock test requirements (see 6.2.1).

4.6.6.2.1 Before and after shock test, the pump and driver and other components susceptible to internal derangement shall be disassembled to the extent necessary and the critical dimensions and running clearances measured, calculated, and recorded. During this disassembly, the critical components and assemblies subject to shock damage and derangement shall be identified and listed in the inspection record and after completion of the test. Shafts, impellers, and turbine rotors, motor rotors, and reduction gears shall be inspected by one of the applicable non-destructive test procedures, other than radiography, specified in accordance with MIL-STD-271.

4.6.6.2.2 Before and after the shock test, tests in accordance with 4.6.3 shall be performed to determine the changes in performance characteristics of the pump. Vibration measurements shall be taken at the bearing caps or housings of the pump and driver at equal speeds during the initial and final capacity test to determine the changes in mechanical operation.

4.6.6.2.3 The unit shall be mounted on the shock machine or barge essentially identical to the actual shipboard installation. The contracting agency will furnish the contractor a drawing of the shipboard mounting arrangement and foundation's stiffness. Horizontal pumps, when tested in the inclined position on the medium weight stock machine, shall be oriented so that the direction of shock is perpendicular to the axis of the pump rotation. The pump shall be in operation during the first, third, and fifth blows of the shock test on the floating barge. Variable speed pumps shall be operated at the minimum speed and pressure required to ensure lubrication of bearings and wearing parts. Other pumps shall be operated at the lowest rated speed.

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4.6.6.2.4 The pump shall be carefully observed during each shock blow and thoroughly examined visually after each blow. After each blow, the unit shall be operated at as close to maximum rated speed as possible and checked for abnormal noises and vibrations and proper functioning of controls. Turbine driven pumps may be air driven. Tightening of bolts during shock tests will not be permitted. If any bolt loosens during the test, the equipment manufacturer shall provide a corrective procedure which must be approved by the design review agent.

4.6.6.2.5 Shock test acceptance criteria is as follows:

(a) There shall be no breakage of parts, including mounting bolts.
(b) There shall be no distortion or derangement of any part which would render the unit incapable of performing as specified,
(c) The amplitude of vibration after test at maximum rated speed shall be less than twice the amplitude measured at the same speed before the test.
(d) Adequate lubrication to all bearings shall be maintained.
(e) Critical dimensions and running clearances have been maintained.
(f) There shall be no significant change in the head-capacity curve.

4.6.6.2.6 Post shock test procedure. The shock tested unit, if it is to be supplied under a contract or order, shall be restored to the as-new condition by replacement of all parts damaged or distorted beyond the as-new design tolerances, Rolling contact bearings shall be replaced regardless of condition. The shock tested rolling contact bearings shall be destroyed. The restored unit shall successfully pass the tests specified in 4.6.1, 4.6.3(a), and 4.6,4, if applicable, Quality conformance test documentation shall certify that the unit was subjected to the shock test and subsequently restored, tested, and inspected in accordance with this specification. A completed parts examination check list shall be supplied (see 6 .2.2) and shall identify the parts which were replaced (such as the bearings) and shall certify that the unit fully conforms to the specifications for unrestricted service.

4.6.6.2.7 When specified (see 6.2.1), pump units shall be shock tested with drivers. Flexible coupled pumps shock tested with one driver will not be required to be shock tested again when supplied with a different driver of equal or less weight. Prime movers are subject to shock tests in accordance with applicable equipment specifications.

4.6.6.3 Inclined operation test. Vertical pumps shall be operated for not less than 30 minutes inclined at an angle from the normal equal to the combination of the maximum permanent list and trim as specified (see 6.2.1). Horizontal pumps shall be tested in accordance with the inclined operation test specified for the driving unit. Operation in the inclined position shall be as close to rated speed and capacity as practicable, The performance point shall be indicated in the test report. The pump shall meet specified performance requirements (see 3.2) without damage while operating in the inclined position.

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4.6.6.4 Vibration test. When specified (see 6.2.1), one pump of a quantity of identical pumps being supplied shall successfully undergo a vibration test in accordance with the requirements of MIL-STD-167-1, type I, and as supplemented in the contract or order. The vibration test need not be repeated on subsequent contracts or orders for pumps of identical design to those previously tested, provided the previous tests included the frequencies specified (see 6.2.1).

4.6.6.4.1 The unit shall be mounted on typical shipboard foundations during the vibration test or the shipboard mounting arrangement shall be simulated in spring mass characteristics except where this mounting arrangement causes the largest test table capacity to be exceeded. Inability to vibration test the unit because of excessive weight or size shall not release the contractor from furnishing equipment which can withstand the specified vibration inputs. Vibration test acceptance criteria shall be in accordance with MIL-STD-167-1, type I.

4.6.6.5 Design evaluation test documentation. The test reports for the design evaluation tests shall be prepared in accordance with accepted engineering practice. The test arrangement and procedure, the test events, and test instruments used, the measured data, and the effects, results, and observations shall be accurate and complete and shall be presented in a professional manner and in a usable durable form to the design review agency for approval. The report shall include a definite statement regarding conformance to the acceptance criteria specified herein and suitability of the unit for its intended application. After approval, test reports shall be distributed as specified in 6.2.2.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6).

5.1 Technical data. Unless otherwise specified (see 6.2.1), technical data depicting the preservation and packaging of data shall be prepared by the contractor.

5.2 Preservation packaging, packing and marking. Pump units and accessories shall be preserved-packaged level A or C, packaged level A, B, or C as specified, (see 6.2.1) and marked in accordance with MIL-P-16789. The shipping containers or method of packing shall comply with National Motor Freight Classification or Uniform Freight Classification Rating, Rules and Regulations or other carrier rules as applicable to the mode of transportation.

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6. NOTES

6.1 Intended use. Centrifugal pumps are for miscellaneous services on board Naval ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) General design requirements:
  (1) Space and weight limitations.
  (2) List, pitch, roll, trim and other ship attitude conditions (see 3.2.1 and 4.6.7.3).
  (3) Environmental vibration frequencies (see 3.2.8.4).
  (4) Ambient temperature, normal, maximum and minimum.
(d) Performance characteristics:
  (1) Rated capacity and range of capacity.
  (2) Rated total head at rated capacity.
  (3) Fluid to be pumped.
  (4) Specific gravity of fluid.
  (5) Temperature of fluid, normal, maximum and minimum.
  (6) Suction head available, normal, maximum and minimum.
  (7) Casting design and test pressures.
  (8) Special performance requirements, if any.
(e) Valves, flanges, fittings and bolting requirements (see 3.2.3 and 3.2.8.6).
(f) Type of driver (see 3.2.4, 3.2.5.1 and 3.2.6.1).
(g) Turbines for driving pumps requirements (see 3.2.5.1).
(h) Applicable specification for motors (see 3.2.6,1).
(i) Class of motor insulation required (see 3.2,6.4).
(j) Motor controller characteristics (see 3.2.7.1).
(k) Shock test requirements (see 3.2.8.1, 3.2.8.2, 4.6.7.2 and 4.6.7.2.7).
(l) Bedplates for close-coupled units (see 3.3.1.4.2),
(m) Noise test, if required and noise level limits (see 3.3.1.6.2 and 4.6.4).
(n) Gauge connections (see 3.3.1.8.8).
(o) Suction and discharge connections (see 3.3.1.9.1).
(p) Whether columns constructed of uniform lengths of pipe shall exceed 5 feet (see 3.3.2.6.2).
(q) Whether shafts shall be furnished in 5 foot sections (see 3.3.2.7.1).
(r) Number of sets of special tools (see 3.6).
(s) Drawings required (see 3.8).
(t) Manuals required and whether complete information for electrical equipment is required (see 3 9.1 and 3.9.3).
(u) Inspection system (see 4.1.3).
(v) Mercury prohibition (see 4.1.5).
(w) Vibration test, if required or other than as specified (see 4.6.6.4).

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(x) When packaging technical data is not required (see 5.1).
(y) Levels of preservation-packaging and packing required (see 5.2).
(z) Ship type and fluid system in which units are to be installed.
(aa) Designation of manufacturer (pump or driver) responsible to the purchaser for comparability of pump and driver and operation of the assembled units.

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.8	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - Contractor Drawing No. - Contractor Delivery of hard copy - Contracting Activity
3.9.1	Manual, technical preliminary	DI-M-2043	Type I of MIL-M-15071
3.9.1	Manual, technical standard	DI-M-2044	Type I of MIL-M-15071
3.9.6	Technical/maintenance overhaul and repair standards (TRSS)	UDI-L-26481	----
4.1.3	Inspection system program plan	DI-R-4803	----
4.3.1	First article inspection report	DI-T-4902	----
4.4 and 4.6	Procedures, test	UDI-T-23732	----
4.4 and 4.6	Reports, test	DI-T-2072	----

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., Vol. I, AMSDL Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2,2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. As used herein "contracting activity" refers to the activity purchasing the equipment covered by this specification. The contracting activity may be the Government or a shipbuilder. The "design review activity" refers to the activity responsible for controlling the technical requirements for design and testing of the equipment. The design review activity is generally a Government command or agency such as NAVSEA, or an authorized representative, Communication with the design review agency should be handled through the contracting activity.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5.2 On board. On board repair parts of the types and quantities shown in table III shall be furnished as specified in the contract or order, as applicable.

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TABLE III. On board repair parts.

Name of part	Quantity per set
Ball or roller bearings. Coupling bushings, pins and other wearing parts. Packing. Mechanical seals. Sleeve bearings.	100 percent complete replacement for all installed units per vessel.
Undersize casing wearing rings. Diaphragm bushings. Stuffing box throat bushings.	One to 4 units per vessel - 100 percent replacement for one pump. Over 4 units per vessel - 100 percent replacement for two pumps.
Rotors, completely assembled, including shaft, impellers, standard size impeller and casing wearing rings, casing diaphragms and bushings, sleeves, keys, nuts, flinger, and complete couplings. Diffusion vanes.	100 percent replacement for one pump.

6.5.2.1 The pump rotor assembly shall be completely assembled ready for installation in the unit for which intended and shall include any nonrotating parts which can be installed only by removing some rotating part or parts of the pump rotor.

6.5.2.2 Pump rotor assembles for type II pumps are not required.

6.5.2.2.1 The quantities of individual repair parts shall not be reduced because of duplication of parts furnished to make the rotor assembly except that standard casing wearing rings shall not be required.

6.5.2.3 Casing wearing rings, (see table III) shall be machined undersize on the internal diameter by at least 0.050 inch in order to permit finish machining as required for replacement aboard ship.

6.5.2.4 Each box containing on board repair parts shall contain a list entitled “List of on board repair parts and tools”. The list shall be in a format suitable for use on outline drawings and in the instruction books. It shall not be less than nominal 8-1/2 by 11 inches in size.

6.5.2.4 1 The format of the list shall include a heading and columns of data for the items listed. The heading shall include titles and applicable entries as follows:

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Reproduced from drawing number _____
Number of ships _____
Application _____
Contract number _____
Manufacturer _____
Quantities are for _____ units per ship
The columns shall include:
Piece number _____
Name of part _____
Quantity _____
Drawing number _____
National stock number _____
Additional columns may be used as applicable.

6.5.2.4.2 The list shall be so treated as to be resistant to oil, water and fading.

6.5.2.5 Stock. The selection, stock numbering and quantities of repair parts for stock shall be determined and processed in accordance with MIL-P-15137

6.5.2.5.1 The cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania, unless otherwise specified in the contract or order.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Design drawings may be submitted to the design review agency for approval.

6.8 A copy of the completed part examination check list for the completed pump, identified by manufacturer's serial number, should be furnished to the contracting activity in duplicate for each pump. On these copies the contractor's quality control manager should certify that the inspection requirements specified herein have been fulfilled and that the unit is in accordance with the approved drawings and the applicable specifications.

6.9 A copy of the completed part examination check list for the completed pump, identified by manufacturer's serial number, should be furnished, On these copies, the contractor's quality control manager should certify that the inspection requirements specified herein have been fulfilled and that the unit is in accordance with the drawings and the applicable specifications.

6.10 The proposed test procedure for this test should be submitted to the contracting agency approximately at the time of drawing submittal.

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6.11 Subject term (key word) listing.

Centrifugal pumps
Impellers
Oil service pumps
Pumps
Stuffing boxes
Water service pumps

6.12 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4320-N248)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-P-18144A(SH)	PUMPS, CENTRIFUGAL, CARGO OIL SERVICE NAVAL SHIPBOARD USE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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